PYTHON PROGRAMMING



21 Days Challenge

Day-17

Day-17

FILE HANDLING - READ/WRITE FILES, WORK WITH JSON FILES AND OTHER FILE OPERATIONS



Abdullah Al Zubaer



Instructor,
Artificial Intelligence & Data Science
Gobeshona Learning Academy.
B.Sc. in Internet of Things and Robotics Engineering
Gazipur Digital University, Gazipur, Bangladesh

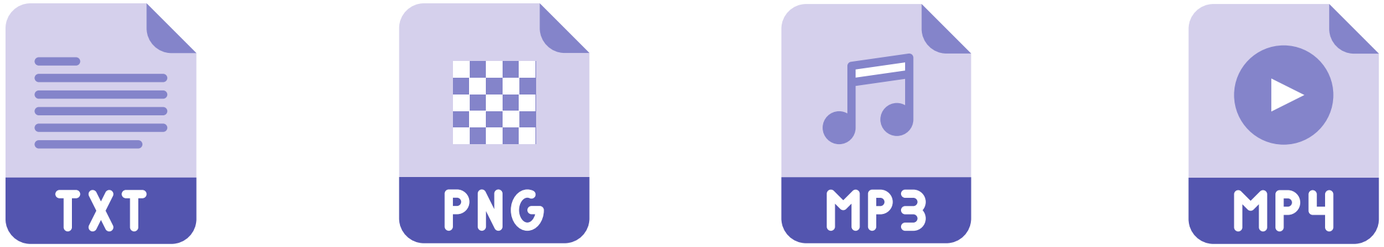
Table Of Content

- What is a File
- Why File Handling
- How to Create a File
- How to Open a File
- File Modes
- How to Read a File
- How to Write a File
- How to Close a File

What is a File

A file is a place to store data permanently on disk.

- Text
- Images
- Audio
- Vedio



Why File Handling

- To store data permanently.
- To read existing data.
- To write new data.

How to Open a File

```
file = open("example.txt", "r")
```

File Modes

- "r" → read (file must exist)
- "w" → write (overwrite or create new file)
- "a" → append (add data at end)
- "+" → read & write
- "x" → create new file, error if exists

How to Create a File

```
file = open("data.txt", "x/w/a")
```

- Create the file if it does not already exist.
- Raise an error (FileExistsError) if the file already exists.

How to Read a File

- read() – Reads the entire file.
- readline() – Reads one line at a time.
- readlines – Reads all lines into a list.

```
file = open("example.txt", "r")
print(file.read())
```

How to Write a File

Overwrites old content to File

```
file = open("example.txt", "w")
file.write("Hello World!")
```

Add data end at old content to File

```
f = open("notes.txt", "a")
f.write("This is a new line!\n")
```

How to Close a File

```
f = open("notes.txt", "r")
print(f.read())
f.close()
```

How to Remove a File

```
import os
os.remove("notes.txt")
```

```
if os.path.exists("notes.txt"):
    os.remove("notes.txt")
```

Best Practice

Automatically closes file.

```
with open("notes.txt", "r") as f:
    print(f.read())
```

Handling Exceptions

```
try:
    file = open("example.txt", "w")
    file.write("This is an example with exception handling.")
finally:
    file.close()
    print ("File closed successfully!!")
```

Summery

- File → Stores data permanently.
- Why File Handling → Save, read, update, or share data.
- Open File → open("file.txt", "mode")
- Modes → "r","w","a" ,"x" ,"+"
- Read Methods → read(), readline(), readlines()
- Write Methods → write(), writelines()
- Close File → close() (always release file)
- with open() → safer, auto-closes file
- Remove File → os.remove("file.txt")



ASSIGNMENT

Create a new file using ‘x' mode, write some text with ‘w', append more text with “a", and then read the content using read(), readline(), and readlines(). Also, use writelines() to add multiple lines, practice with open() for safe handling, and finally remove the file using os.remove().

QUESTION SESSION

THANK YOU

https://gobeshona.com.bd/